

$(r-1, c-1)$ idx - cols - 1	$(r-1, c)$ idx - cols	$(r-1, c+1)$ idx - cols + 1
$(r, c-1)$ idx - 1	(r, c) — this thread idx = row*cols + col	$(r, c+1)$ idx + 1
$(r+1, c-1)$ idx + cols - 1	$(r+1, c)$ idx + cols	$(r+1, c+1)$ idx + cols + 1

Blue = orthogonal neighbours: horizontal = idx +/- 1, vertical = idx +/- cols (a whole row away). Grey = diagonals (both).